RIEMANN-ROCH AND DUALITY FOR LOCALLY FREE SHEAVES ON A GORENSTEIN CURVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the invertibility step, the self-applied Riemann-Roch computation of deg(omega_X), the substituted chi formula, and the duality symmetry are all retained verbatim.

Editorial note. The label lemma-rr names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Gorenstein implies CM and the dualizing module. Duality for Schemes, Lemma duality-lemma-gorenstein-CM, Section duality-section-gorenstein; Same chapter, Lemma lemma-duality-dim-1-CM. Makes omega_X a dualizing module and, by the Gorenstein property, an invertible sheaf.
- The Riemann-Roch relation and Euler characteristics. Same chapter, Equation equation-rr, Equation equation-degree-c1 and Lemma lemma-euler. Supplies chi(E) in terms of the Chern class and chi(O_X), the identity chi(omega_X) = - deg(omega_X), and the base Euler characteristic computation.
- Duality for locally free sheaves. Same chapter, Remark remark-rework-duality-locally-free. Gives the symmetry dim H^i(E) = dim H^{1-i}(E^vee (+) omega_X) for locally free E.

Source reference. Original lemma 0C00: sources/stacks-chow/tags.txt, source lines 15070–15070. Prerequisite pointer invoked at proof line(s) 1243; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0DWG: sources/stacks-chow/tags.txt, source lines 17396–17396. Prerequisite pointer invoked at proof line(s) 1239; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0AWV: sources/stacks-chow/tags.txt, source lines 13733–13733. Prerequisite pointer invoked at proof line(s) 1248; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original equation 0BRY: sources/stacks-curves/pinned-original-source.tex, source lines 1127–1127. Prerequisite pointer invoked at proof line(s)

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1258; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original equation 0BRZ: sources/stacks-curves/pinned-original-source.tex, source lines 1134–1134. Prerequisite pointer invoked at proof line(s) 1249; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0BS2: sources/stacks-curves/pinned-original-source.tex, source lines 796–796. Prerequisite pointer invoked at proof line(s) 1221; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0BS3: sources/stacks-curves/pinned-original-source.tex, source lines 864–864. Prerequisite pointer invoked at proof line(s) 1245; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0BS5: sources/stacks-curves/pinned-original-source.tex, source lines 1174–1174. Prerequisite pointer invoked at proof line(s) 1254; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0BS4: sources/stacks-curves/pinned-original-source.tex, source lines 893–893. Prerequisite pointer invoked at proof line(s) 1261; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1 (Riemann-Roch)**.** *Let X be a proper scheme over a field k which is Gorenstein and equidimensional of dimension 1. Let ω_X be as in Lemma 0BS2. Then*

1. *ω_X is an invertible $\mathcal{O}_X$-module,*
2. *$\deg(\omega_X) = -2\chi(X, \mathcal{O}_X)$,*
3. *for a locally free $\mathcal{O}_X$-module $\mathcal{E}$ of constant rank we have*

$$\chi(X, \mathcal{E}) = \deg(\mathcal{E}) - \tfrac{1}{2} rank(\mathcal{E}) \deg(\omega_X)$$

and $\dim_k(H^i(X, \mathcal{E})) = \dim_k(H^{1-i}(X, \mathcal{E}^\vee \otimes_{\mathcal{O}_X} \omega_X))$ for all $i \in \mathbf{Z}$.

Proof. Recall that Gorenstein schemes are Cohen-Macaulay (Duality for Schemes, Lemma 0C00) and hence ω_X is a dualizing module on X, see Lemma 0BS3. It follows more or less from the definition of the Gorenstein property that the dualizing sheaf is invertible, see Duality for Schemes, Section 0AWV. By (0BRZ) applied to ω_X we have

$$\chi(X, \omega_X) = \deg(c_1(\omega_X) \cap [X]_1) + \chi(X, \mathcal{O}_X)$$

Combined with Lemma 0BS5 this gives

$$2\chi(X, \mathcal{O}_X) = -\deg(c_1(\omega_X) \cap [X]_1) = -\deg(\omega_X)$$

the second equality by (0BRY). Putting this back into (0BRZ) for $\mathcal{E}$ gives the displayed formula of the lemma. The symmetry in dimensions is a consequence of duality for X, see Remark 0BS4. □